

Prompt for Kiro

You are creating the **PRISM Doubly Robust Modeling README** for the PRISM project.

You have access to the following documents:

- PRISM_Doubly_Robust_Modeling_Workflow.ipynb
- PRISM_Doubly_Robust_Engineering_Specification.md
- PRISM_Causal_Forest_Modeling_README.md

The Causal Forest README is the design template.

The goal is **NOT** to invent a new report structure.

The goal is to create a Doubly Robust README that reads as though it was written by the same author, using the same formatting conventions, same writing style, same section organization, same level of detail, and same visualization style.

Primary Objective

Use the Causal Forest README as the structural blueprint.

Replace the causal forest methodology with the Doubly Robust methodology while preserving the overall flow of the document.

Someone reading the two READMEs side-by-side should immediately recognize that they belong to the same project.

Maintain Identical Overall Structure

Preserve the same hierarchy whenever applicable:

- Background
- Business Question
- Project Objectives
- Analytical Task 1
- Analytical Task 2
- Evaluation Roadmap

- Level summaries
- Analytical Tasks 3–7
- Supporting files
- Final summaries

Only modify section names where necessary because of methodological differences.

Match Writing Style

The writing should closely match the Causal Forest README.

Specifically:

- same paragraph lengths
- same tone
- same level of technical detail
- same balance of technical and business explanation
- same markdown formatting
- same table formatting
- same use of equations
- same use of explanatory callouts
- same narrative transitions

Do **not** introduce a different writing style.

Mirror Tables Whenever Possible

If a section in the causal forest README contains a table that also makes sense for Doubly Robust, recreate the same table layout.

Examples include:

- data review summary
- event count summary

- true benefit validation
- decile summaries
- overlap summaries
- business value tables
- supporting file sections
- evaluation roadmap
- level summaries

Only change the contents to reflect the Doubly Robust outputs.

Mirror Figures Whenever Possible

Whenever a chart or visualization exists in the Causal Forest README and there is an equivalent output from the Doubly Robust notebook:

use

- the same chart type
- the same titles
- the same ordering
- the same captions
- the same placement
- the same markdown formatting

Examples include:

- distribution plots
- average benefit by decile
- risk tier vs benefit group
- cumulative gross savings
- marginal savings
- SHAP plots

- feature importance plots

These should look like the Doubly Robust counterpart of the causal forest figures.

Preserve Mermaid Diagram Style

If the causal forest README contains a workflow diagram that has an equivalent Doubly Robust version,

create a corresponding Mermaid diagram using the same style.

Only modify the workflow to reflect the Doubly Robust estimation procedure.

Preserve Evaluation Philosophy

The evaluation philosophy should remain consistent.

The README should answer questions such as:

- Are the treatment effects credible?
- How well does the model recover synthetic truth?
- Does the model identify meaningful high-benefit members?
- Can the results be explained?
- Does targeting improve business value?

The methods differ.

The evaluation philosophy should remain the same.

Preserve Analytical Tasks

Maintain the same progression whenever appropriate.

For example:

Task 1

- Explain the modeling framework.

Task 2

- Review the data.

Task 3

- Model credibility / diagnostics.

Task 4

- Treatment-effect analysis.

Task 5

- Decile and subgroup analysis.

Task 6

- Explainability.

Task 7

- Business value.

Only modify sections where the Doubly Robust methodology requires something different.

Keep Existing Doubly Robust Components

The following sections are unique to the Doubly Robust workflow and should remain.

Examples include:

- ForestDRLearner methodology
- doubly robust estimation equations
- orthogonalization explanation
- nuisance models
- doubly robust pseudo-outcomes
- cross-fitting
- doubly robust treatment-effect estimation
- any diagnostics unique to ForestDRLearner

Do **not** force these into the causal forest wording.

Keep them technically correct.

Preserve Unique Doubly Robust Outputs

If the notebook generates outputs unique to Doubly Robust, retain them.

These should not be rewritten to imitate the causal forest README.

Instead,

integrate them naturally into the existing report structure.

SHAP Section

Use the same organization as the Causal Forest README.

Include:

- global SHAP importance
- signed SHAP contribution table
- SHAP explanation
- interpretation
- comparison with true synthetic drivers

Maintain the same style and layout.

Business Value Section

Follow the exact organizational style of the Causal Forest README.

Include:

- cumulative targeting comparison
- marginal targeting advantage
- ROI assumptions
- interpretation
- supporting files

Only substitute the Doubly Robust results.

Supporting Files

Match the formatting used throughout the Causal Forest README.

Every analytical task should conclude with supporting output files where applicable.

Formatting Requirements

Maintain:

- identical heading hierarchy
- markdown style
- code block style
- equation formatting
- AUTO_TABLE tags
- AUTO_CHART tags
- AUTO_TEXT tags
- supporting file formatting

Do not change naming conventions without necessity.

Deliverable

Produce a polished README that:

- feels like the companion document to the Causal Forest README,
- uses the Doubly Robust notebook as the technical source,
- follows the engineering specification,
- preserves the PRISM documentation style,
- is publication quality,
- and requires minimal editing before inclusion in the PRISM repository.